

Asad Ikram

MSc Data Analytics (Distinction) · Chevening Scholar 2024/25 · Emotion and sentiment measurement in online text about policy and brands
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PROFILE

Over seven years I have worked for 5+ companies and 300+ freelance clients, mostly collecting and analysing online data. At Dubizzle Labs, leading a team that collected text in four languages across 15 countries, I saw that people write differently on forums and on social platforms, yet their text was treated as comparable. As a Chevening Scholar I took that problem to the University of Portsmouth, into an MSc dissertation on public reaction to UK economic policy and a methods manuscript with Dr Ella Haig, then into sentiment and emotion coding for brands at M+C Saatchi Fluency and into ArtemisAI, the company I founded, where I found my own system's accuracy figure rising partly because its reference had moved. I now want to study the emotional appeals and cues in chatbot output written for politicians and civil servants, with measures that are tested against human coders before they are believed.

Fit with A.I. Feels (SP1)	SP1 asks how emotions are expressed in chatbot output for policymakers. I already detect emotion in text at scale, with the NRC lexicon, a fine-tuned DistilBERT emotion classifier and LLM coding on a 24-emotion taxonomy, and I have seen two LLM annotators agree less often on emotion labels than on sentiment. My manuscript counts a result only if it holds under two measurement families and treats the human coder as the criterion; the same rules would govern emotion coding of LLM output. My Python pipelines could collect responses to a structured set of policy prompts across models and versions, and I would draft the codebook from a close reading of a purposive subsample, piloted by two trained coders.
Research interests	Emotional appeals and cues in AI-generated text on political and policy issues, and whether chatbot output reproduces or flattens the emotional registers of political communication; what emotion-sensitive AI means for policymaking and democracy; validity of automated emotion measurement across models, prompts, platforms and time; combining large-scale computational coding with close reading of what emotion words do in a text.

ANALYSIS METHODS

Emotion measurement	NRC emotion lexicon (eight basic emotions and two sentiment categories; proportional scores aggregated over time and set against predicted sentiment); binary emotion presence indicators for invariance and differential item functioning tests; a 24-emotion multi-label taxonomy (9 positive or approach, 10 negative or avoidance, 5 transitional) coded per text by LLM; a fine-tuned DistilBERT emotion classifier in production; comparing measures, since a single lexicon can mislead. Human validation to date covers sentiment (Cohen's $\kappa \approx 0.78$); emotion is lexicon- or LLM-coded so far.
Content analysis	Automated content analysis validated against human coding; codebook design, piloting and revision; sentiment, emotion, need-state, toxicity, topic and intent coding; topic modelling (BERTopic) to find candidate themes; LLM-assisted annotation checked by hand; multilingual text translated before coding. Fine-tuned encoder classifiers (BERT, XLM-R, DistilBERT, DeBERTa-v3).
Statistics	Inter-coder reliability (Cohen's κ); agreement between labelling tiers versus criterion validity; McNemar tests, bootstrap confidence intervals, chi-square tests; probability calibration; simulation with known ground truth; checks of coverage, denominators and whether groups can be pooled; cross-market comparison and indexing; minimum-base and small-cell rules; sensitivity analyses.
Measurement	Generalizability theory, measurement invariance, differential item functioning and score linking, specified in a protocol and tested by simulation; not yet fitted to collected data.
Research practice	Ethics approval and GDPR-compliant handling of public text; analysis anchors fixed in advance, before any counts were computed; documented, reproducible analysis pipelines.
Software	Python, PyTorch, SQL, PySpark; R (working knowledge). Data access through official APIs and licensed feeds (Meta Graph, Reddit, YouTube, Brandwatch, Sprinklr); AWS, Airflow, Docker.

EDUCATION

2024 - 2025 awarded Dec 2025	MSc Data Analytics, Distinction. University of Portsmouth, United Kingdom. Credit-weighted average 70.5/100 (Distinction threshold 70). Master's project 78/100, supervised by Dr Ella Haig. Modules included Intelligent Data and Text Analytics, Big Data and Machine Learning, and Explainable AI. 180 UK credits (conventionally 90 ECTS), taught and assessed in English.
2017 - 2021 awarded Jan 2022	BSc Computer Science. National University of Computer and Emerging Sciences (FAST-NUCES), Pakistan. Four years full time, 130 credit hours completed. CGPA 2.75 on a 0.00 to 4.00 scale.

RESEARCH

2025
Master's project

Cross-Platform Sentiment Analysis of Public Reaction to UK Economic Policies.

- **Finding.** The same policy event produced different sentiment and emotion profiles on different platforms, so a pooled index assumed, without testing, that the classifier measures the same construct everywhere.
- **Study design.** Ethics-approved quantitative study (Portsmouth TETHIC-2025-111094) of about 279,000 public posts and comments on UK economic policy from X/Twitter, Reddit, YouTube and Quora, collected through official APIs and licensed feeds under GDPR-compliant procedures. Human-annotated validation set of 200 posts balanced across platforms (reported inter-annotator agreement Cohen's $\kappa \approx 0.78$).
- **Classification and testing.** Fine-tuned BERT classifier compared with BiLSTM by McNemar test ($p < 0.001$) and bootstrap intervals, alongside logistic regression and VADER baselines, with probability calibration. Chi-square tests of sentiment by policy topic (taxation and fiscal policy, cost of living, welfare, energy). A stated design goal: policymakers must understand the basis of sentiment ratings.
- **Emotion.** NRC lexicon counts of eight emotions and two sentiment categories for each post, aggregated over time as proportional scores and compared with predicted sentiment: anger and fear co-occurred with negative sentiment, joy and anticipation with positive, trust with neutral.
- **Correction.** Re-auditing for the manuscript, I found the headline macro-F1 of 0.878 was agreement with the lexicon-seeded training labels, not with the human set. The manuscript says so and specifies a larger stratified human criterion with a second annotator.

2026
manuscript

Ikram, A. and Haig, E. "Is the Platform Part of the Measurement? A Protocol and Simulation Study for Cross-Platform Equivalence of Machine-Labelled Policy Sentiment." Complete; not published and not submitted.

- Treats each social platform as a survey mode and asks whether a classifier measures the same construct on each: a generalizability study against a human criterion, invariance and differential item functioning with platform as the grouping facet, and score linking.
- 500-replication simulation with known ground truth: an index that pools platform scores at face value misstates a known event shift by roughly an eighth and reports the wrong sign of aggregate sentiment on nearly a quarter of days; the measurement-adjusted index gets the sign wrong on about one day in sixty. The paper frames this as the error an analyst would pass on to a policymaker.
- Emotion enters as eight binary NRC presence indicators in invariance and differential item functioning models grouped by platform. NRC results are labelled lexicon-conditional, and non-invariance counts only if it appears under both classifier families (after Czarnek and Stillwell, 2022). Models specified, not yet fitted to data.

No peer-reviewed publications to date.

EXPERIENCE

Feb 2025 - present

Founder and Chief Technology Officer, ArtemisAI Ltd (trading as ViralData), United Kingdom (remote). Social media analytics for small businesses, running in production ahead of a December 2026 launch.

- **Agreement is not validity.** The two labelling tiers agree on 82.9% of sentiment, 74.4% of emotion and 0.0% of toxicity labels (a taxonomy mismatch), while the toxicity model reports 96.7% average confidence: a precise, confident number with no criterion behind it.
- **A moving reference.** Reported average accuracy rose from 64.7 (v1) to 85.8 (v5), but v1 to v3 were scored against the human holdout and v4 onward against Claude's labels, so part of the gain is the reference moving rather than the model improving. The emotion classifier's v5 score of 84 is also against Claude's labels.
- **Products.** Flop detection (warns when a page risks an underperforming post), crisis detection from toxicity scores, best posting times, post drafting, and a collaboration feature built on network theory (a networkx affinity graph of pages).
- **Models.** Six fine-tuned encoder classifiers code 111,000+ posts and 3.39 million comments from 30+ public pages (Meta Graph API): XLM-R (sentiment), DistilBERT (emotion), DeBERTa-v3 (topic, intent), Detoxify (toxicity) and a language classifier. They learn from a tiered labelling chain via APIs: we hand-labelled about 500 posts and comments and matched Claude and DeepSeek against them; Claude then labels about 6,000 posts, and rules mined from its reasoning prompt DeepSeek to label about 12,000 more.
- **Images and video.** A PyTorch fusion classifier over CLIP embeddings of image, OCR text and caption (macro-F1 0.93 on a 149-item gold set; production labels from Claude Haiku 4.5 vision); a hybrid OCR pipeline (token-F1 0.928); Whisper transcription.
- **Evaluation design.** Release gate of macro-F1 above 0.85 on both a 97-post human holdout and a 4,000-item stratified test set, then a human check of 200 random posts; weekly KL-divergence drift alarms.

- **Research base.** Features reviewed against the literature: Grimmer, Roberts and Stewart's *Text as Data* (hold back future data for testing, never present a prediction as a cause, check labels against human judges), published popularity-prediction benchmarks for flop detection, and the ranking code X released as open source.

Dec 2024 - present

Data Engineer, consultant, M+C Saatchi Fluency, United Kingdom (remote). Analysing online data (social posts, forums, reviews, search) for brands, and supporting UK Government social listening pipelines.

- **Method under challenge.** CFMOTO 2027 launch report (eight brands): eight-theme sentiment codebook, composite metrics with written formulas and an emotional profile of each brand against the competitor average. With the team, I answered client challenges with source-mix audits, a consumer-base minimum for all brands, re-tagging agreement checks and consumer-only and non-US sensitivity runs; anything the checks did not support was reported as directional.
- **Social discourse analysis.** US women's soccer: over 100,000 consumer posts filtered in Python from a year of Twitter, forum and Reddit data; accounts classified by speaker; BERTopic topics merged by hand into a codebook; every post LLM-coded for sentiment and themes, hand-checked on a 500-post pilot.
- **Emotion and need-state analysis.** Abbott: about 170,000 multilingual posts and reviews on infant formula from 12 platforms in Asia, Latin America and the Middle East, coded in LLM-assisted stages (language detection and translation, relevance, need states, a 24-emotion multi-label taxonomy with one flag per emotion, 18 themes) and compared across sources and markets with small-cell suppression, since the mix of sources shapes an emotion profile. Reckitt: over 33,000 US reviews and about 19,000 search keywords LLM-coded on one 16-need framework; a survey deep dive from respondent-level SPSS data on the filters and bases of a validated earlier deck.
- **Social listening pipelines** on AWS for UK Government departments, Amazon, Ford and Nike. UK Government: contributed to a reporting pipeline (ingest, AI tagging, comment processing, data-quality monitoring). Amazon: Sprinklr ETL (cleaning, deduplication, translation, tagging into Postgres and Tableau). Ford: 68 collectors on owner forums and other sources across ten platforms, with a self-healing repair agent.

Feb 2022 - Aug 2024

Senior Data Engineer, Dubizzle Labs (Dubizzle, Bayut, Zameen, OLX), Pakistan. Promoted twice in two years.

- **Competitor analysis.** Analysed e-commerce and real estate stores and online discussions via web scraping and the data warehouse I redesigned, for dashboards predicting revenue and showing what customers say.
- **Collection.** Led three engineers and 500+ collectors gathering text in English, Arabic, Urdu and Hindi from 15 MENA countries, where I first saw forum and social text treated as comparable.

2018 - present

Earlier and concurrent engagements

- **Fix.com,** Canada, part-time since May 2023 (50M+ data points, 200+ sources); **VendueTech,** Ireland, Lead Data Engineer, 2025; **Prefe,** Italy, 2024; **CXG,** Dubai, 2025 to 2026; **freelance,** 2018 to 2026, 300+ clients.

AWARDS, PUBLIC WORK AND LANGUAGES

Chevening	Chevening Scholar 2024/25 (UK FCDO), selected from 70,000+ applicants in 160 countries; funded the MSc.
Public work	Author of web-scraping-guide.com, a free public reference on web data collection, its ethics and data quality.
Languages	Urdu (native). English: MSc in English; IELTS Academic 6.5. No Dutch yet; I would begin learning it on arrival.
Availability	Can start on 1 January 2027 and relocate to within commuting distance of Amsterdam; would take on the teaching (about 10%) and project work.

References available on request.